AquaTwin

Measured filter health and failure prediction for smart water purifiers

Track: 2, Water Purifier	**Area:** Prognostics of filter failure using data analytics, combined with input water quality sensing beyond TDS

1 Problem and Idea

Purifier filters do not wear out at a fixed rate. A sediment cartridge can clog within weeks after a few tanker deliveries, an RO membrane on hard borewell water scales much faster than one on municipal supply, and a shifted O-ring can let untreated water slip past the membrane overnight. Most purifiers, smart ones included, still estimate filter life from litres dispensed and time elapsed, sometimes weighted by input TDS. That figure is calculated, not measured, so cartridges get replaced while still usable or kept running after performance has dropped. On a product sold with zero running cost for two years, both mistakes are expensive: wasted cartridges on one side, complaints and unplanned service visits on the other.

> **AquaTwin** is a software layer that runs on the purifier controller and in the cloud. During each purification cycle it reads signals the purifier already produces: input and output TDS, water temperature, pump run time, valve and mode state, and the time the 8 L tank takes to fill, which gives permeate flow without an extra sensor. From these it computes temperature-corrected health indicators for each stage and tracks membrane fouling with a Kalman filter. A cloud model projects this forward using the household's usage pattern and current water source, and reports remaining life as a range (for example, 38 days, likely between 30 and 47) along with the probable cause. The app shows one clear status per filter with a guided replacement, and the service team sees the same forecast, so the right cartridge reaches the home before water quality is affected. An optional low-cost sensor kit improves accuracy on premium models.

Why two areas. Source water is the biggest factor in how fast filters degrade, and many Indian homes switch between municipal, tanker and borewell supply. AquaTwin recognises these switches from the combined pattern of TDS level, TDS variation and temperature, with no new sensor. Without this, the failure forecast would be wrong every time the source changes.

Baseline vs target

Metric	Today (litre / time based)	AquaTwin target
Replacement trigger	Fixed litres or months	Measured health of each stage
Remaining-life error, 30 days before end of life	No forecast available	15% with sensor kit, 25% without
Warning before output TDS or flow goes out of spec	None, user notices first	At least 14 days ahead
Detection of RO bypass or O-ring leak	Only if user checks TDS	Within 3 purification cycles
Cause of decline identified	Not identified	85% accuracy across 6 fault types
Cartridges replaced while still usable	To be measured in Round 2	Reduced by at least 25%
Added hardware cost per unit	Not applicable	Rs. 0, or under Rs. 600 with kit

Targets will be validated on our test rig before Round 2.

What sets it apart

- Health is measured from how each stage actually performs, so two homes with the same litre count can get different and correct answers.
- Each stage gets its own forecast and a likely cause, because a clogged pre-filter and a leaking O-ring need different fixes.
- The base version adds no hardware and can reach existing purifiers through an over-the-air update.
- Units share only fitted model parameters, never raw data, so new installations start with forecasts tuned to their pincode and water source.

2 Technical Feasibility

The core method is adapted from normalised performance monitoring, the standard practice in industrial RO plants (ASTM D4516). Raw permeate flow and salt rejection shift with temperature, pressure and feed salinity, so they are corrected to reference conditions before being trended. A sustained 10 to 15% drop in normalised permeate flow, or a

similar rise in salt passage or pressure drop, is the accepted sign that a membrane needs attention. The physics is well established; our contribution is running it at household scale, mostly on signals the purifier already has, and adding a probabilistic forecast on top.

$$TCF = \exp[2640\ (1/298 - 1/(273 + T))] \qquad NPF = Q_p\ (NDP_{ref} / NDP) / TCF$$

$$\text{Salt passage} = TDS_{out} / TDS_{in} \qquad J = NDP / [\mu(T)\ (R_m + R_f(t))]$$

Each cycle, the edge firmware keeps only steady-state readings (pump running for over 60 s with stable flow) and updates a three-state Extended Kalman Filter: fouling resistance R_f, its growth rate, and the TDS sensor offset. In the cloud, a particle filter of about 500 samples runs once a day, simulating future fouling under the last 14 days of household usage until the end-of-life threshold is reached; the spread of results gives the P10 to P90 range. A gradient-boosted classifier reads the shape of the change to name the cause, and Bayesian change-point detection on feed TDS and temperature flags a change of water source.

Fault	What AquaTwin looks for	Sensing
Sediment clog	Tank fill time and inlet pressure drop rise, salt passage unchanged	Existing (kit adds dP)
RO scaling or fouling	Gradual fall in normalised flow and rise in salt passage	Existing
RO bypass or O-ring leak	Sudden jump in salt passage within one cycle	Existing
Pump wear	More current drawn for the same pressure and flow	Sensor kit
UV lamp ageing	Lamp current drift combined with burn hours and switching count	Kit, or hours only
Carbon exhaustion	Estimated from volume and expected chlorine for the detected source	Existing (estimate)

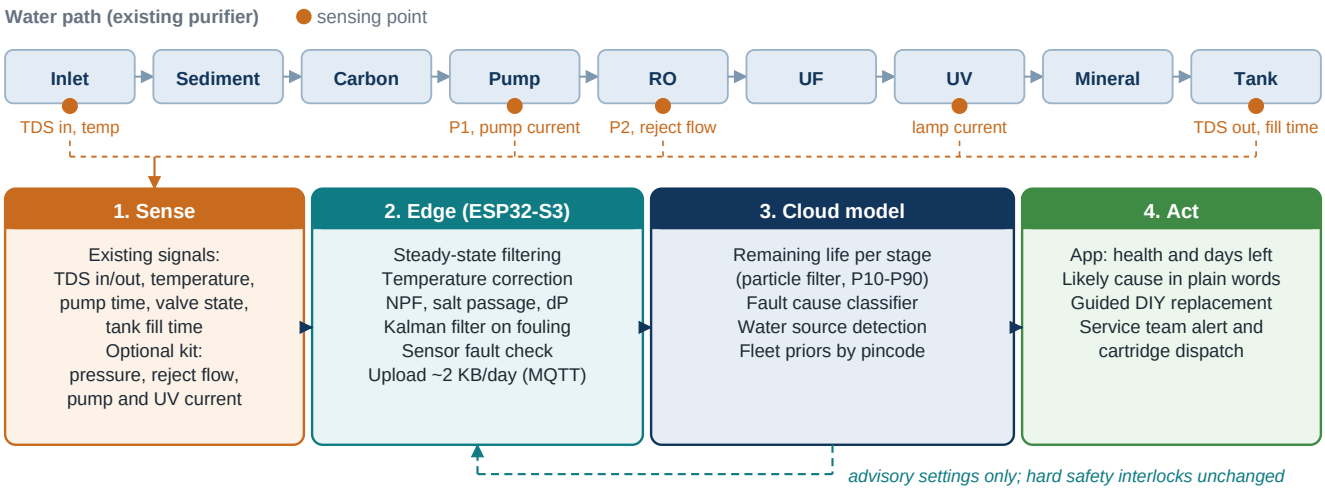
Key specifications. Domestic booster pumps deliver about 0.4 to 0.7 MPa, so the optional pressure sensors are 0 to 1.2 MPa, ±1% full scale. An 8 L tank filling in 25 to 40 minutes gives a flow estimate repeatable to about ±3% per cycle, fine enough to catch a 10% decline well before the user notices. Sampling is 1 Hz while the pump runs, summarised to roughly 2 KB per day. The three-state filter is light work for an ESP32-S3 (240 MHz, 512 KB SRAM), and a reduced on-device model keeps alerts working offline.

Training data. Membranes take months to foul, so we combine three sources: a physics-based simulator calibrated on our rig, used cartridges of known age collected from local RO technicians, and controlled faults (partly blocked sediment cartridges, a deliberately damaged O-ring, and hardness-dosed feed water to speed up scaling).

Safety. The model is advisory. Existing interlocks (output TDS cut-off, UV fault lockout, leak and dry-run protection) stay in firmware and cannot be overridden, and AquaTwin can only recommend replacement earlier than the manufacturer's maximum, never later. If a reading is stuck, out of range or inconsistent, the unit falls back to the existing litre and time counter. Added sensors run at 3.3 to 5 V from the isolated supply, inline parts use food-grade push-fit fittings rated above pump pressure, and only cycle summaries leave the device, with user consent under the DPDP Act, 2023. Hardness dosing is used only on the test rig, never on drinking water.

3 Execution Feasibility

System block diagram



Components, cost and sourcing

Subsystem	Key components	Approx. cost	Sourcing
Host purifier	Domestic RO + UV purifier with float switch and TDS sensing	Rs. 8,000-12,000	Retail
Controller and sensing	ESP32-S3, 2 TDS probes with temperature compensation, DS18B20, 5 V supply	Rs. 2,000	Local
Optional sensor kit	2 pressure transducers (0 to 1.2 MPa), low-flow sensor, 2 INA219 current sensors, food-grade fittings	Rs. 2,500	Imported, 1 to 2 weeks
Test rig	Spare and used RO membranes, sediment cartridges, O-rings, 20 L feed tank	Rs. 3,000	Local technicians
Software	ESP-IDF, MQTT, FastAPI, TimescaleDB, Python, Flutter	Free	Open source

The full prototype costs roughly **Rs. 15,000 to 20,000**, most of it the host purifier. At production volume the sensor kit is targeted under Rs. 600 and the base version at zero added cost. The only sourcing risk is the imported pressure and low-flow sensors; we will order from two vendors in the first week, and the base version does not need them. The only custom part is a 3D-printed sensor bracket, made in-house.

Build plan to Round 2

Week	Work	Done when
1	Buy purifier and sensors, tap existing signals, set up data logging to the cloud, first version of the simulator	Live data from the rig
2	Calibrate sensors, collect used cartridges, build health indicators and the Kalman filter	Flow repeatable within 3%
3	Run fault tests (aged membrane, clogged sediment, O-ring bypass, source switch); build forecast and cause models	First accuracy results
4	Build the app screens, compare base version against sensor kit, rehearse the demo	Full demo runs end to end

Live demo. We swap a healthy RO membrane for a field-aged one, and within three cycles the app changes from healthy to fouling, about 40 days left. We then loosen an O-ring to show the bypass alert within a single cycle, and switch the feed from low to high TDS water to show the forecast shorten as the source changes.